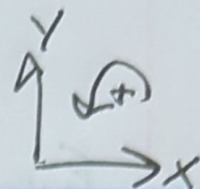
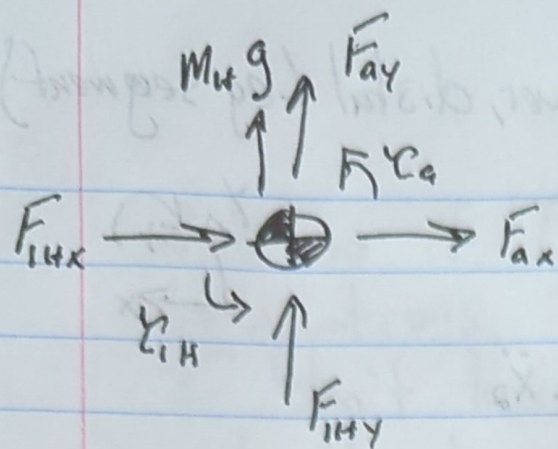
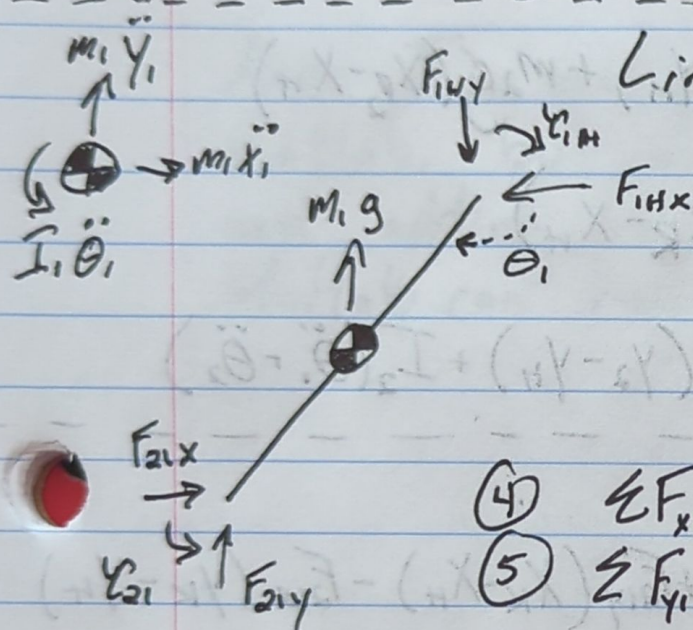


Hip (Floating Base)



$$\begin{aligned} \textcircled{1} \quad \sum F_{xH} &= F_{ax} + F_{Hx} & \textcircled{1} &= m_H \ddot{X}_H \\ \textcircled{2} \quad \sum F_{yH} &= F_{ay} + F_{Hy} + M_H g & &= m_H \ddot{Y}_H \\ \textcircled{3} \quad \sum M_H &= \tau_a + \tau_{iH} & &= I_H \ddot{\theta}_H \quad (\theta_H \text{ and } \theta_i \text{ are same}) \end{aligned}$$



θ_1 is angle of leg segment w.r.t. x-axis. Shown at about -135° here.

$$\begin{aligned} \textcircled{4} \quad \sum F_{x_1} &= F_{21x} - F_{Hx} & &= m_1 \ddot{X}_1 \\ \textcircled{5} \quad \sum F_{y_1} &= F_{21y} - F_{Hy} + m_1 g & &= m_1 \ddot{Y}_1 \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad \sum M_1 &= \tau_{21} - \tau_{iH} + m_1 g (x_1 - x_H) + F_{21y} (x_k - x_H) - F_{21x} (y_k - y_H) \\ &= m_1 \ddot{Y}_1 (x_1 - x_H) - m_1 \ddot{X}_1 (y_1 - y_H) + I_1 \ddot{\theta}_1 \end{aligned}$$

$$\begin{aligned} X_1 &= X_H + \frac{L_1}{2} \cos \theta_1 & Y_1 &= Y_H + \sin \theta_1 \frac{L_1}{2} \\ X_k &= X_H + L_1 \cos \theta_1 & Y_k &= Y_H + L_1 \sin \theta_1 \end{aligned}$$

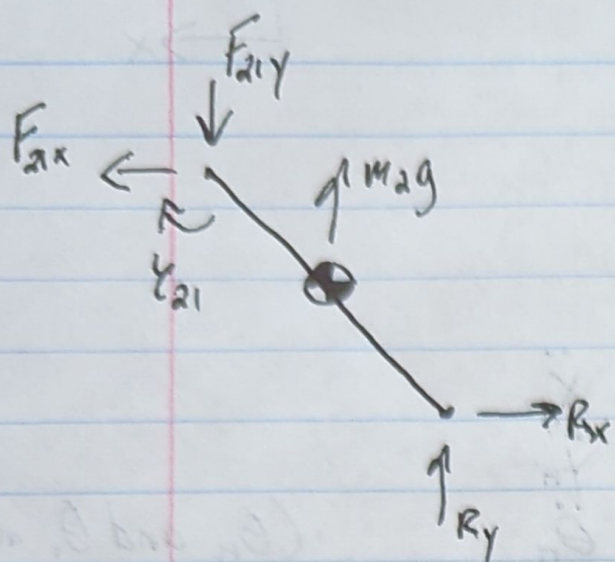
$\textcircled{10} = \textcircled{1} + \textcircled{4}$ to isolate F_{21x} : $F_{ax} + F_{21x} = m_H \ddot{X}_H + m_1 \ddot{X}_1 \Rightarrow F_{21x} = m_H \ddot{X}_H + m_1 \ddot{X}_1 - F_{ax}$

$\textcircled{11} = \textcircled{2} + \textcircled{5}$ to isolate F_{21y} : $F_{ay} + m_H g + F_{21y} + m_1 g = m_H \ddot{Y}_H + m_1 \ddot{Y}_1$

$$\Rightarrow F_{21y} = m_H \ddot{Y}_H + m_1 \ddot{Y}_1 - F_{ay} - (m_H + m_1)g$$

$\textcircled{12}$ And: $\tau_{21} = +K_2 (\theta_2 - \theta_{20})$

Link 2 (lower, distal Leg segment)



$$m_2 \ddot{y}_2$$

$$I_2 (\ddot{\theta}_2 + \ddot{\theta}_1) \rightarrow m_2 \ddot{x}_2$$

$$\begin{aligned} \textcircled{7} \quad \sum F_x &= R_x - F_{21x} = m_2 \ddot{x}_2 \\ \textcircled{8} \quad \sum F_y &= R_y - F_{21y} + m_2 g = m_2 \ddot{y}_2 \end{aligned}$$

$$\begin{aligned} \textcircled{9} \quad \sum M &= -\tau_{21} + R_y (x_F - x_H) - R_x (y_F - y_H) + m_2 g (x_{c_2} - x_H) \\ &\quad + F_{21x} (y_K - y_H) - F_{21y} (x_K - x_H) \end{aligned}$$

$$= m_2 \ddot{y}_2 (x_2 - x_H) - m_2 \ddot{x}_2 (y_2 - y_H) + I_2 (\ddot{\theta}_1 + \ddot{\theta}_2)$$

$$\textcircled{13} = \textcircled{3} + \textcircled{6} = \tau_a + \tau_{21} + m_1 g (x_1 - x_H) + F_{21y} (x_K - x_H) - F_{21x} (y_K - y_H)$$

$$\textcircled{\star} = I_H \ddot{\theta}_1 + I_2 \ddot{\theta}_2 + m_1 \ddot{y}_1 (x_1 - x_H) - m_1 \ddot{x}_1 (y_1 - y_H)$$

Plug in $\textcircled{10}, \textcircled{11}, \textcircled{12}$ and now $\textcircled{13}$ is a state equation with only state variables in it.

State variables being $x_H, y_H, \theta_1, \theta_2$

*Note: I could separate the hip and Link 1 by the hip spring, and then make $\tau_{1H} = K_1 (\theta_1 - \theta_{10})$ and then τ_a would be truly an input. But I don't think I want to. Instead I'll try to make τ_a the torque from the hip spring. and $F_{ax} = F_{ay} = 0$.

$$\tau_a = +K_1 (\theta_1 - \theta_{10})$$

Ok so if I use eq (13), (7), (8), and (9) that's 4 equations. I have 4 variables. But! I don't know what R_x, R_y are so I actually have 6 variables. So if I add in the GRF spring eqs again like I've done in the past then I can have 6 eqs and 6 variables. Should I try it?

And I'm going to try plugging eq (13) into matlab, but also just doing (1) through (9) and adding the GRF eqs and seeing if Matlab can handle it.